

## Project – Interactive Dashboard with Power Query and Power Pivot

As part of this project, I built a comprehensive, interactive sales and shipping dashboard in Excel by following the video tutorial. The dashboard development process included data importation, transformation, modeling, visualization, and dynamic interaction through slicers and sparklines - entirely within Excel.

### Part 1: Getting and Shaping Data with Power Query

#### 1. Setting Up the Dashboard Sheet:

I opened a new Excel workbook and renamed one sheet as **"Dashboard"**, which served as the canvas for all final visuals and interactions.

#### 2. Importing Data from a Folder:

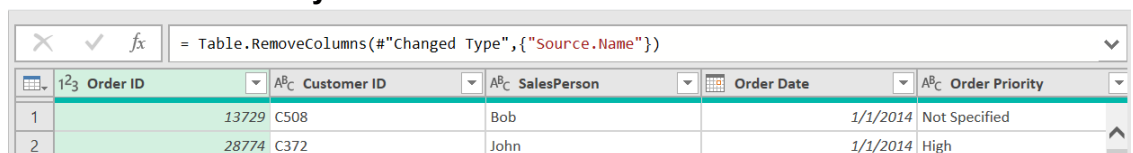
I navigated to **Data > Get Data > From File > From Folder**, pasted the folder path containing multiple CSV files, and confirmed the import.

#### 3. Combining and Editing Data:

From the preview dialog, I selected **Combine & Edit** to open the **Power Query Editor**. Power Query automatically identified the structure and presented the combined data in an editable view.

#### 4. Data Cleaning and Transformation in Power Query:

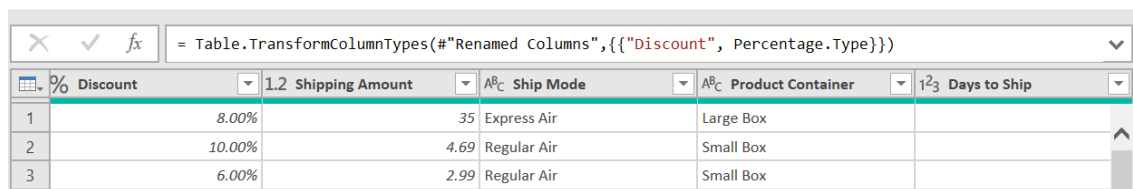
a) **Removed unnecessary columns** such as Source.Name.



The screenshot shows the Power Query Editor with the formula bar containing the M code: `= Table.RemoveColumns("#Changed Type",{ "Source.Name" })`. The data table below has columns: Order ID, Customer ID, SalesPerson, Order Date, and Order Priority.

|   | Order ID | Customer ID | SalesPerson | Order Date | Order Priority |
|---|----------|-------------|-------------|------------|----------------|
| 1 | 13729    | C508        | Bob         | 1/1/2014   | Not Specified  |
| 2 | 28774    | C372        | John        | 1/1/2014   | High           |

b) **Verified and set data types** appropriately to ensure compatibility with Power Pivot.

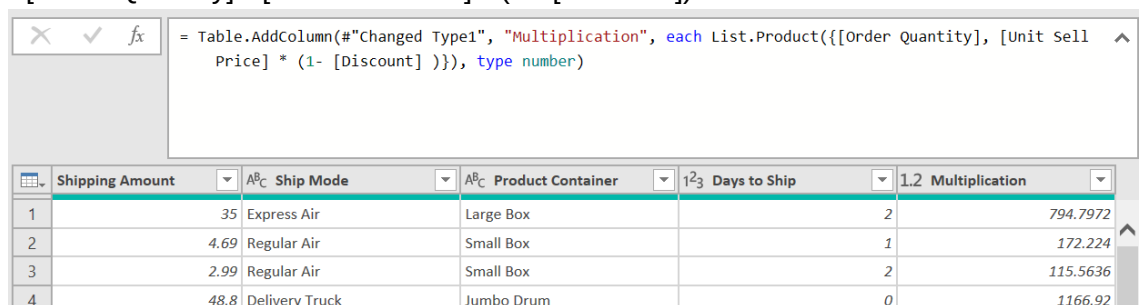


The screenshot shows the Power Query Editor with the formula bar containing the M code: `= Table.TransformColumnTypes("#Renamed Columns",{{ "Discount", Percentage.Type}})`. The data table below has columns: Discount, Shipping Amount, Ship Mode, Product Container, and Days to Ship.

|   | Discount | Shipping Amount | Ship Mode   | Product Container | Days to Ship |
|---|----------|-----------------|-------------|-------------------|--------------|
| 1 | 8.00%    | 35              | Express Air | Large Box         |              |
| 2 | 10.00%   | 4.69            | Regular Air | Small Box         |              |
| 3 | 6.00%    | 2.99            | Regular Air | Small Box         |              |

c) **Calculated "Sale Amount"** using a custom formula:

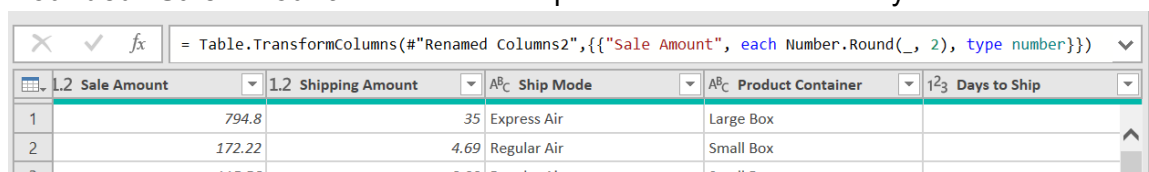
`= [Order Quantity] * [Unit Sale Price] * (1 - [Discount])`.



The screenshot shows the Power Query Editor with the formula bar containing the M code: `= Table.AddColumn("#Changed Type1", "Multiplication", each List.Product({[Order Quantity], [Unit Sell Price] * (1 - [Discount])}), type number)`. The data table below has columns: Shipping Amount, Ship Mode, Product Container, Days to Ship, and Multiplication.

|   | Shipping Amount | Ship Mode      | Product Container | Days to Ship | Multiplication |
|---|-----------------|----------------|-------------------|--------------|----------------|
| 1 | 35              | Express Air    | Large Box         | 2            | 794.7972       |
| 2 | 4.69            | Regular Air    | Small Box         | 1            | 172.224        |
| 3 | 2.99            | Regular Air    | Small Box         | 2            | 115.5636       |
| 4 | 48.8            | Delivery Truck | Jumbo Drum        | 0            | 1166.92        |

d) **Rounded "Sale Amount"** to two decimal places for better readability.



The screenshot shows the Power Query Editor with the formula bar containing the M code: `= Table.TransformColumns("#Renamed Columns2",{{ "Sale Amount", each Number.Round(_, 2), type number}})`. The data table below has columns: Sale Amount, Shipping Amount, Ship Mode, Product Container, and Days to Ship.

|   | Sale Amount | Shipping Amount | Ship Mode   | Product Container | Days to Ship |
|---|-------------|-----------------|-------------|-------------------|--------------|
| 1 | 794.8       | 35              | Express Air | Large Box         |              |
| 2 | 172.22      | 4.69            | Regular Air | Small Box         |              |
| 3 | 115.56      | 2.99            | Regular Air | Small Box         |              |

e) **Created "Days to Ship"** by subtracting Order Date from Ship Date using **Date > Subtract Days**.

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= Table.AddColumn("#Reordered Columns", "Subtraction", each Duration.Days([Ship Date] - [Order Date]), Int64.Type)

|   | Discount | 1.2 Shipping Amount | A <sup>8</sup> <sub>C</sub> Ship Mode | A <sup>8</sup> <sub>C</sub> Product Container | 1 <sup>2</sup> <sub>3</sub> Subtraction |
|---|----------|---------------------|---------------------------------------|-----------------------------------------------|-----------------------------------------|
| 1 | 0.08     | 35                  | Express Air                           | Large Box                                     | 2                                       |
| 2 | 0.1      | 4.69                | Regular Air                           | Small Box                                     | 1                                       |

### 5. Naming and Loading Data:

Renamed the query to **"Sales Data"** and then Loaded the data using **Close & Load To...**, selecting **Only Create Connection** and **Add this data to the Data Model**.

## Part 2: Building the Data Model with Power Pivot

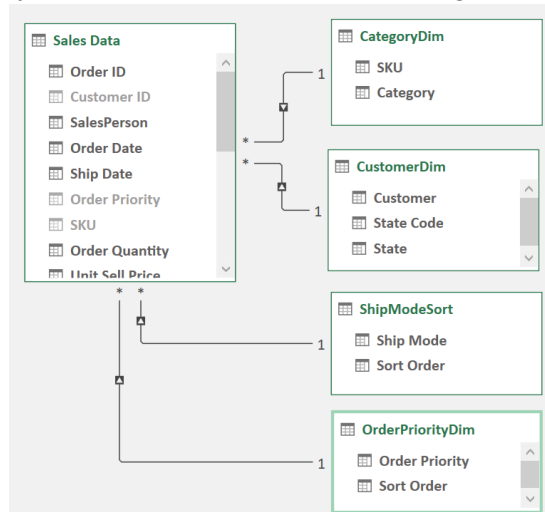
### 1. Adding Dimension Tables:

I added supporting tables (e.g., Customers, Product Categories, Shipping Mode, Order Priority) to the Data Model using **Power Pivot > Add to Data Model**.

### 2. Creating Relationships:

In **Power Pivot > Diagram View**, I linked:

- SKU** from "Sales Data" to the **Category** table.
- Customer ID** to the **Customers** table.
- Shipping** and **priority** fields to their respective mapping tables.



### 3. Model Configuration:

- Hide key fields from the "Sales Data" table after relationships were established to avoid redundancy.
- Set custom sort orders (e.g., for Shipping Mode: Delivery Truck → Regular Air → Express Air) using Sort by Column.
- Formatted fields for reporting: Sale Amount and Shipping Amount as Currency, Discount as Percentage.

### 4. Creating DAX Measures and Calculated Columns:

Added meaningful calculations for enhanced insights:

- Average Shipping Price per Item: `=DIVIDE(SUM('Sales Data'[Shipping Amount]),SUM('Sales Data'[Order Quantity]),"`
- `=FORMAT([Order Date], "yyyy")` for extracting year.
- `=CONCATENATE("Qtr", INT((MONTH([Order Date]) + 2) / 3))` for quarter names.
- `=FORMAT([Order Date], "MMM")` for abbreviated month names.

| Order Date ...                                   |         |              |           |        |          |             |       |             |            |            |               |                  |
|--------------------------------------------------|---------|--------------|-----------|--------|----------|-------------|-------|-------------|------------|------------|---------------|------------------|
| =CONCATENATE("Qtr",INT((MONTH(Order Date)+2)/3)) |         |              |           |        |          |             |       |             |            |            |               |                  |
| ...                                              | Or...   | Order ...    | Unit S... | Dis... | Sale ... | Shipping... | S...  | Product ... | Days ...   | Order D... | Order Date... | Order Date (M... |
| 1                                                | 3/20... | Low          | SK...     | 46     | \$6.48   | 9.00%       | \$271 | \$8.73      | Regular... | Small Box  | 2 2014        | Qtr1             |
| 2                                                | 15/2... | High         | SK...     | 6      | \$6.48   | 0.00%       | \$39  | \$10.05     | Regular... | Small Box  | 2 2014        | Qtr1             |
| 3                                                | 3/20... | Not Speci... | SK...     | 16     | \$6.48   | 5.00%       | \$99  | \$6.86      | Regular... | Small Box  | 2 2014        | Qtr1             |
| 4                                                | 20/2... | Medium       | SK...     | 7      | \$6.48   | 7.00%       | \$42  | \$9.68      | Regular... | Small Box  | 2 2014        | Qtr1             |
| 5                                                | 3/20... | Critical     | SK...     | 49     | \$6.48   | 1.00%       | \$314 | \$7.37      | Regular... | Small Box  | 2 2014        | Qtr2             |
| 6                                                | 22/2... | Not Speci... | SK...     | 50     | \$6.48   | 0.00%       | \$324 | \$5.11      | Regular... | Small Box  | 2 2014        | Qtr2             |
| 7                                                | 31/2... | Medium       | SK...     | 5      | \$6.48   | 2.00%       | \$32  | \$7.91      | Regular... | Small Box  | 2 2014        | Qtr2             |

## Part 3: Creating PivotTables, PivotCharts, and Visual Components

### 1. Creating PivotTables and Charts:

- Built PivotTables for each required metric, placing each on a separate sheet and renaming them for clarity (e.g., Dimension Tables, Map Pivot).

|    | A      | B          | C | D        | E          | F              | G | H              | I          | J | K              | L          |
|----|--------|------------|---|----------|------------|----------------|---|----------------|------------|---|----------------|------------|
| 1  | SKU    | Category   |   | Customer | State Code | State          |   | Ship Mode      | Sort Order |   | Order Priority | Sort Order |
| 2  | SKU001 | Bikes      |   | C001     | VA         | Virginia       |   | Delivery Truck | 1          |   | Critical       | 1          |
| 3  | SKU002 | Components |   | C002     | NJ         | New Jersey     |   | Express Air    | 3          |   | High           | 2          |
| 4  | SKU003 | Bikes      |   | C003     | NY         | New York       |   | Regular Air    | 2          |   | Low            | 4          |
| 5  | SKU004 | Bikes      |   | C004     | IA         | Iowa           |   |                |            |   | Medium         | 3          |
| 6  | SKU005 | Components |   | C005     | LA         | Louisiana      |   |                |            |   | Not Specified  | 5          |
| 7  | SKU006 | Bikes      |   | C006     | MI         | Michigan       |   |                |            |   |                |            |
| 8  | SKU007 | Components |   | C007     | IL         | Illinois       |   |                |            |   |                |            |
| 9  | SKU008 | Bikes      |   | C008     | NM         | New Mexico     |   |                |            |   |                |            |
| 10 | SKU009 | Bikes      |   | C009     | IA         | Iowa           |   |                |            |   |                |            |
| 11 | SKU010 | Components |   | C010     | LA         | Louisiana      |   |                |            |   |                |            |
| 12 | SKU011 | Bikes      |   | C011     | OK         | Oklahoma       |   |                |            |   |                |            |
| 13 | SKU012 | Components |   | C012     | NC         | North Carolina |   |                |            |   |                |            |
| 14 | SKU013 | Components |   | C013     | OK         | Oklahoma       |   |                |            |   |                |            |
| 15 | SKU014 | Components |   | C014     | UT         | Utah           |   |                |            |   |                |            |
| 16 | SKU015 | Components |   | C015     | OK         | Oklahoma       |   |                |            |   |                |            |
| 17 | SKU016 | Components |   | C016     | MD         | Maryland       |   |                |            |   |                |            |
| 18 | SKU017 | Components |   | C017     | AL         | Alabama        |   |                |            |   |                |            |
| 19 | SKU018 | Bikes      |   | C018     | OR         | Oregon         |   |                |            |   |                |            |
| 20 | SKU019 | Components |   | C019     | ID         | Idaho          |   |                |            |   |                |            |
| 21 | SKU020 | Components |   | C020     | MO         | Missouri       |   |                |            |   |                |            |
| 22 | SKU021 | Components |   | C021     | VT         | Vermont        |   |                |            |   |                |            |
| 23 | SKU022 | Bikes      |   | C022     | FL         | Florida        |   |                |            |   |                |            |
| 24 | SKU023 | Components |   | C023     | NC         | North Carolina |   |                |            |   |                |            |
| 25 | SKU024 | Bikes      |   | C024     | RI         | Rhode Island   |   |                |            |   |                |            |
| 26 | SKU025 | Components |   | C025     | WA         | Washington     |   |                |            |   |                |            |

| A              | B                               | C      | D              | E                       | F    | G | H | I | J | K | L |
|----------------|---------------------------------|--------|----------------|-------------------------|------|---|---|---|---|---|---|
| Row Labels     | Average Shipping Price per Item |        | Row Labels     | Average of Days to Ship |      |   |   |   |   |   |   |
| Regular Air    |                                 |        | Critical       |                         |      |   |   |   |   |   |   |
| Wrap Bag       |                                 | \$0.10 | Delivery Truck |                         | 1.49 |   |   |   |   |   |   |
| Jumbo Box      |                                 | \$0.16 | Regular Air    |                         | 1.53 |   |   |   |   |   |   |
| Small Pack     |                                 | \$0.16 | Express Air    |                         | 1.49 |   |   |   |   |   |   |
| Jumbo Drum     |                                 | \$0.19 | High           |                         |      |   |   |   |   |   |   |
| Small Box      |                                 | \$0.28 | Delivery Truck |                         | 1.45 |   |   |   |   |   |   |
| Medium Box     |                                 | \$0.47 | Regular Air    |                         | 1.40 |   |   |   |   |   |   |
| Large Box      |                                 | \$1.30 | Express Air    |                         | 1.44 |   |   |   |   |   |   |
| Express Air    |                                 |        | Medium         |                         |      |   |   |   |   |   |   |
| Wrap Bag       |                                 | \$0.10 | Delivery Truck |                         | 1.58 |   |   |   |   |   |   |
| Small Pack     |                                 | \$0.18 | Regular Air    |                         | 1.46 |   |   |   |   |   |   |
| Small Box      |                                 | \$0.28 | Express Air    |                         | 1.46 |   |   |   |   |   |   |
| Medium Box     |                                 | \$0.56 | Low            |                         |      |   |   |   |   |   |   |
| Large Box      |                                 | \$1.15 | Delivery Truck |                         | 4.00 |   |   |   |   |   |   |
| Delivery Truck |                                 |        | Regular Air    |                         | 4.28 |   |   |   |   |   |   |
| Jumbo Drum     |                                 | \$1.65 | Express Air    |                         | 4.25 |   |   |   |   |   |   |
| Jumbo Box      |                                 | \$1.98 | Not Specified  |                         |      |   |   |   |   |   |   |
| Grand Total *  |                                 | \$0.50 | Delivery Truck |                         | 1.47 |   |   |   |   |   |   |
|                |                                 |        | Regular Air    |                         | 1.47 |   |   |   |   |   |   |
|                |                                 |        | Express Air    |                         | 1.47 |   |   |   |   |   |   |
|                |                                 |        | Grand Total    |                         | 2.03 |   |   |   |   |   |   |

- Created PivotCharts from the PivotTables:
  - Sales by Category (Bar Chart)
  - Sales Over Time by Quarter (Line Chart)
  - Average Shipping Price per Item (Bar Chart)
  - Average Days to Ship (Bar Chart)

### 2. Generating Map Charts:

- Created PivotTables using **State** and either **Total Sales** or **Average Days to Ship**.
- Inserted **Map Charts**.

| Row Labels    | Sum of Sale Amount | Row Labels    | Average of Days to Ship |
|---------------|--------------------|---------------|-------------------------|
| Alabama       | \$393,137          | Alabama       | 1.95                    |
| Arizona       | \$226,229          | Arizona       | 1.79                    |
| California    | \$451,054          | California    | 2.20                    |
| Colorado      | \$318,867          | Colorado      | 1.98                    |
| Connecticut   | \$500,716          | Connecticut   | 2.11                    |
| Delaware      | \$188,041          | Delaware      | 1.87                    |
| Florida       | \$548,641          | Florida       | 2.13                    |
| Georgia       | \$569,311          | Georgia       | 1.87                    |
| Idaho         | \$415,551          | Idaho         | 2.33                    |
| Illinois      | \$268,940          | Illinois      | 1.89                    |
| Indiana       | \$343,631          | Indiana       | 2.02                    |
| Iowa          | \$252,125          | Iowa          | 1.76                    |
| Kansas        | \$463,502          | Kansas        | 2.01                    |
| Kentucky      | \$298,086          | Kentucky      | 2.24                    |
| Louisiana     | \$274,656          | Louisiana     | 1.87                    |
| Maine         | \$398,644          | Maine         | 2.04                    |
| Maryland      | \$316,655          | Maryland      | 2.27                    |
| Massachusetts | \$268,590          | Massachusetts | 1.87                    |

### 3. Creating Sparkline Tables:

- Developed PivotTables that summarized:
  - Sales over time by Category**
  - Top 4 Customers by Sales**
- Created Sparklines using **Insert > Sparklines**.

| 2014      | 2015      | 2016      |
|-----------|-----------|-----------|
| Qtr1      | Qtr2      | Qtr3      |
| \$26,826  | \$21,577  | \$31,424  |
| \$922,444 | \$93,519  | \$730,675 |
| \$9,939   | \$11,366  | \$13,931  |
| \$276,783 | \$243,564 | \$259,350 |

| 2014     | 2015     | 2016     |
|----------|----------|----------|
| Jan      | Feb      | Mar      |
| \$95,760 | \$16,203 | \$10,408 |
| \$313    | \$12,705 | \$14,883 |
|          | \$447    | \$499    |

## Part 4: Assembling and Finalizing the Dashboard

- Placing Visuals:**
  - Cut and pasted all charts and tables to the **Dashboard** sheet.
  - Aligned and resized elements for clear layout.
- Formatting the Dashboard:**
  - Removed chart field buttons, gridlines, unnecessary axes, and legends.
  - Used **Paste Formatting** to apply consistent styles across charts.
- Adding and Configuring Slicers:**
  - Inserted Slicers: **Year, Category, Product Container, Salesperson, and Shipping Mode**.
  - Connected each slicer to the appropriate PivotTables via **Report Connections**.
  - Removed slicer headers and resized slicers for uniform presentation.

